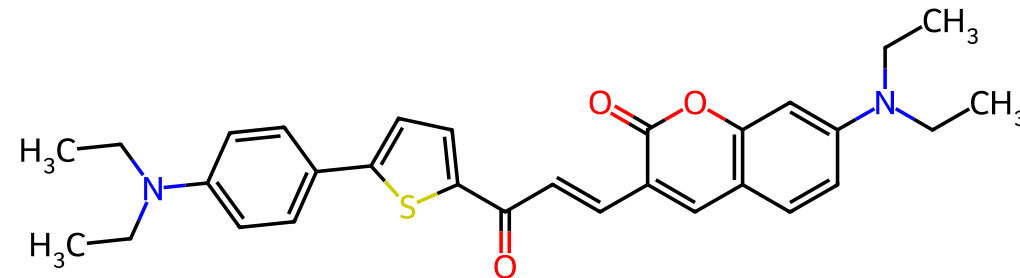


CMR-GOLD-058 | $\log P_{\text{exp}} = 0.97$

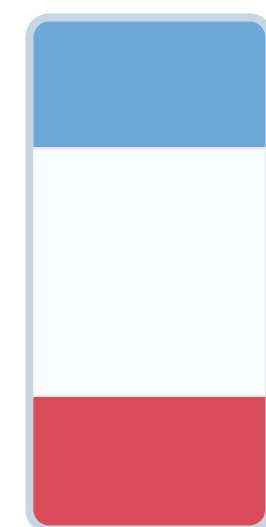
donor- π -acceptor coumarin



$\Delta \log P$ (Exp - Pred)

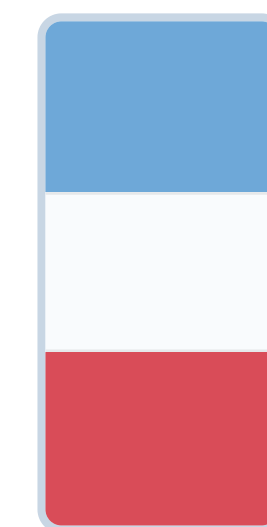
XLOGP3		-6.11
WLOGP		-6.03
Silicos-IT		-6.71
Consensus		-5.19

ELECTRONIC ORIGIN



4.12 eV

CMR-GOLD-043
N = 2 control



2.61 eV

CMR-GOLD-058
N = 2 overestimation

VS

HOMO-LUMO gap · same N count, different
electronic regime

RELIABILITY-TRIAGE FRAMEWORK

FM0	serviceable N-free	-0.26
FM1	polar N = 1–3 high risk	-2.03
FM2	conjugated N, provisional	-0.87
FM3	high-N cancellation	-0.78
FM4	π -extended underestimation	+0.75

FM1 → **experimental $\log P$ priority**

TABLE OF CONTENTS TEXT

Fragment-based and fragment-influenced $\log P$ estimates can systematically overestimate polar nitrogen-containing coumarins, with consensus bias reaching -2.08 log units in the lowest experimental lipophilicity range. A targeted ten-compound DFT/MEP diagnostic panel shows that severe overestimation is consistent with donor- π -acceptor embedding and frontier-orbital softening rather than nitrogen count alone. The resulting FM0-FM4 reliability-triage framework identifies when experimental lipophilicity measurement should take priority.